HDOC DAY 62

Topic: Merging two sorted arrays using pointers

Aim

Merge two ascending-order variable-length integer arrays into a third sorted array, preserving duplicates, using the two-pointer technique and a separate function.

Algorithm

1. Start.
2. Read positive sizes n1 and n2 and the sorted elements of arr1 and arr2.
3. Pass the arrays by address to merge_sorted_arrays().
4. Compare current elements using pointers i and j; append the smaller element to merged and advance its pointer.
5. Copy any remaining elements from either input array.
6. Print the merged array and stop.

Test Case Table

Test	n1, n2	Input arrays	Expected merged output
1	4, 4	1 3 5 7 / 2 4 6 8	1 2 3 4 5 6 7 8
2	4, 3	1 2 2 4 / 2 3 5	1 2 2 2 3 4 5
3	3, 5	1 4 7 / 2 3 5 6 8	1 2 3 4 5 6 7 8
4	3, 2	1 3 5 / 6 7	1 3 5 6 7

C Program

```
#include <stdio.h>

void merge_sorted_arrays(int *arr1, int n1, int *arr2, int n2)
{
    int merged[n1 + n2];
    int i = 0, j = 0, k = 0;

    while (i < n1 && j < n2) {
        if (*(arr1 + i) <= *(arr2 + j))
            merged[k++] = *(arr1 + i++);
        else
            merged[k++] = *(arr2 + j++);
    }
    while (i < n1)
        merged[k++] = *(arr1 + i++);
    while (j < n2)
        merged[k++] = *(arr2 + j++);

    printf("Merged array: ");
    for (i = 0; i < k; i++)
        printf("%d ", merged[i]);
    printf("\n");
}

int main(void)
{
    int n1, n2, i;
```

```
    printf("Enter size of first array: ");
    if (scanf("%d", &n1) != 1 || n1 <= 0) {
        printf("Invalid first array size.\n");
        return 1;
    }
    printf("Enter size of second array: ");
    if (scanf("%d", &n2) != 1 || n2 <= 0) {
        printf("Invalid second array size.\n");
        return 1;
    }
    int arr1[n1], arr2[n2];
    printf("Enter first sorted array elements:\n");
    for (i = 0; i < n1; i++)
        scanf("%d", &arr1[i]);
    printf("Enter second sorted array elements:\n");
    for (i = 0; i < n2; i++)
        scanf("%d", &arr2[i]);
    merge_sorted_arrays(arr1, n1, arr2, n2);
    return 0;
}
```

Compilation and Execution (Mac Terminal)

```
clang -std=c99 day62.c -o day62
./day62
```

Sample Output

```
Enter size of first array: 4
Enter size of second array: 3
Enter first sorted array elements:
1 2 2 4
Enter second sorted array elements:
2 3 5
Merged array: 1 2 2 2 3 4 5
```

Result

The sorted arrays were merged successfully using pointers and a user-defined function. Time complexity: O(n1 + n2).